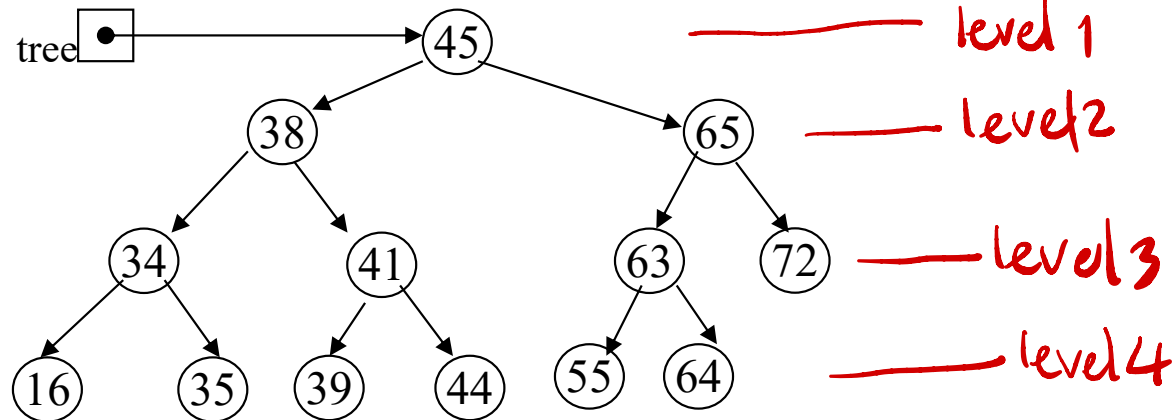


1. Given the following binary tree:



height

1

2

3

4

(a) What is the inorder traversal of the tree? **Left → Root → Right**

16, 34, 35, 38, 39, 41, 44, 45, 55, 63, 64, 65, 72

(b) What is the preorder traversal of the tree? **Root → Left → Right**

45, 38, 34, 16, 35, 41, 39, 44, 65, 63, 55, 64, 72

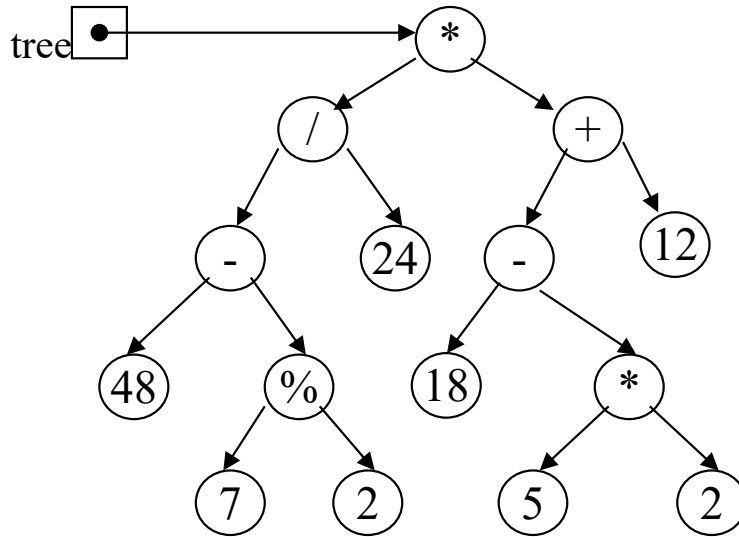
(c) What is the postorder traversal of the tree? **Left → Right → Root**

16, 35, 34, 39, 44, 41, 38, 55, 64, 63, 72, 65, 45

(d) What is the height of the tree? What nodes are on level 2?

height = 4, Level 2 nodes: 38, 65

2. Given the following binary expression tree:



(a) What is the inorder traversal of the tree?

$$[(48 - (7 \cdot 2)) / 24] * [(18 - (5 * 2)) + 12]$$

(b) What is the postorder traversal of the tree?

$$48 \ 7 \ 2 \ \% \ - \ 24 \ / \ 18 \ 5 \ 2 \ * \ - \ 12 \ + \ *$$

(c) What does it evaluate to if using integer division?

20

(d) What does it evaluate to if using float division?

39.1666 ...

c)

$$((48 - 1) / 24) * (18 - 10 + 12)$$

$$(47 / 24) * 20$$

$$1 * 20 = 20$$

$$d) [(48 - 1) / 24] * [18 - 10 + 12]$$

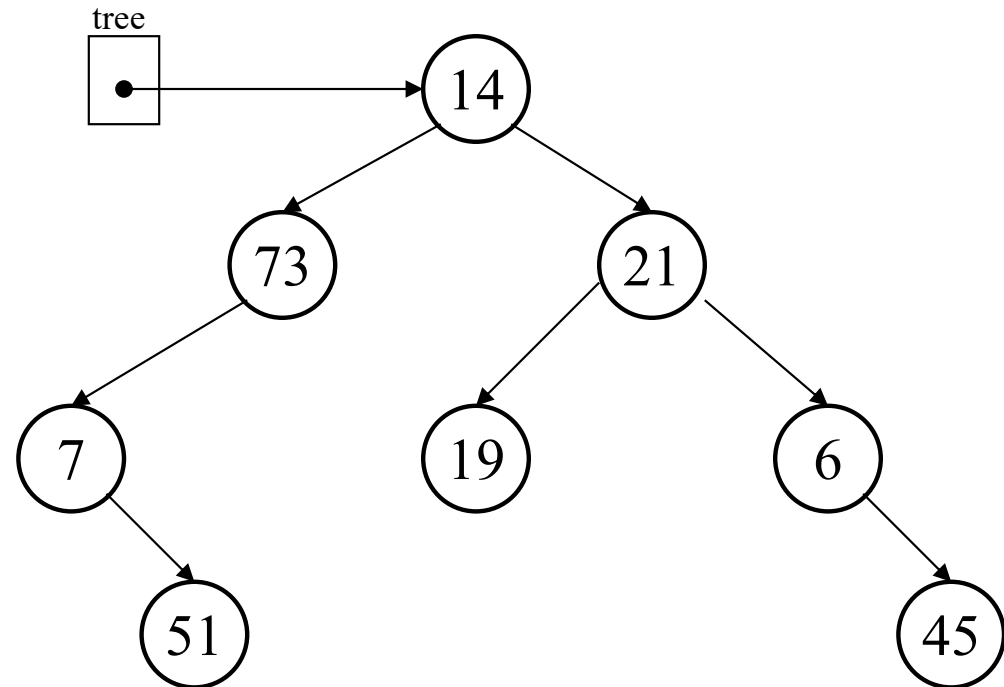
$$= (47 / 24) * 20$$

$$= 1.95833 * 20$$

$$= 39.1666$$

3. The elements in a binary tree area to be stored in an array. Each element is a nonnegative int value.
- What value can you use as a dummy value, if the binary tree is not complete? null
 - Show the contents of the array, given the tree illustrated below

[0]	14
[1]	73
[2]	21
[3]	7
[4]	null
[5]	19
[6]	6
[7]	null
[8]	51
[9]	null
[10]	null
[11]	null
[12]	null
[13]	null
[14]	45



4. Given the array pictured below, draw the binary tree that can be created from its elements.

[0]	35	✓
[1]	20	✓
[2]	71	✓
[3]	40	✓
[4]	52	✓
[5]	63	✓
[6]	null	✓
[7]	17	✓
[8]	25	✓
[9]	null	✓
[10]	7	✓
[11]	null	✓
[12]	45	✓

